Euclid's Elements — Book I
Proposition 8

Formal Reconstruction — Technical Documentation

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Side-side-side angle rigidity

1 Introduction

Proposition I.8 is Euclid's Side–Side–Side rigidity theorem. Given two nondegenerate triangles with their three corresponding sides congruent, the proposition concludes that all three corresponding angles are congruent.

Historically, Euclid proves this by superposition and uses Proposition I.7 to exclude a second possible position of the remaining vertex. The current production theorem `euclid_proposition_8` is much shorter: it calls the reusable theorem `HilbertSSS` and exposes the three angle fields of the returned triangle-congruence result.

The diagrammatic audit cannot stop at that public call. Current source inspection shows that `HilbertSSS` itself contains substantial geometry. In Hilbert's Theorem 18 construction, an auxiliary point is placed on the side of the base opposite the original third vertex, SAS is used once to transfer a side, Hilbert's Theorem 17 supplies an opposite-side rigidity angle, and a second SAS application recovers the complete congruence data. The audited chapter therefore records both the public wrapper and this hidden mathematical engine.

2 The Classical Configuration

Let ABC and DEF be nondegenerate triangles with

$$AB \cong DE, \qquad BC \cong EF, \qquad AC \cong DF.$$

The task is to prove

$$\angle BAC \cong \angle EDF, \qquad \angle ABC \cong \angle DEF, \qquad \angle ACB \cong \angle DFE.$$

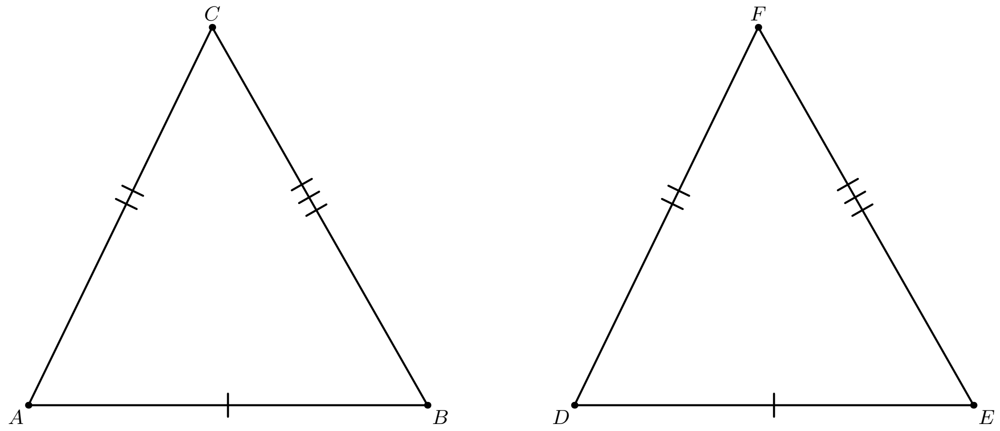

Figure 1: The data of Proposition I.8. The three pairs of corresponding sides of ABC and DEF are congruent.

2.1 Euclid's superposition

Euclid places one triangle on the other so that B coincides with E and C with F. The transported position of A will be denoted by G. If $G \neq D$, then G and D are two distinct points on the same side of the common base EF satisfying

$$EG \cong ED, \qquad FG \cong FD.$$

Proposition I.7 excludes exactly this configuration, so $G = D$.

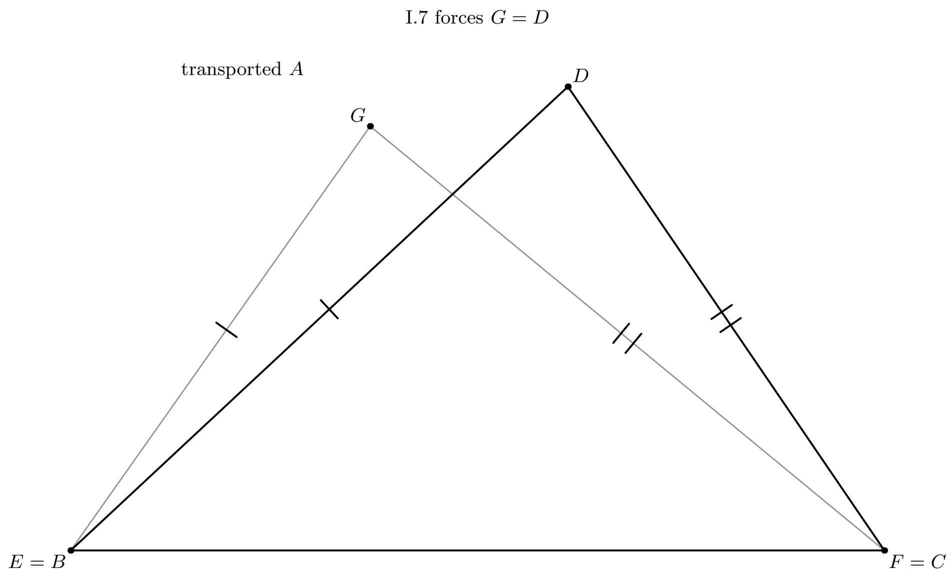

Figure 2: Euclid's superposition step. After the base has been identified, the transported vertex G and the target vertex D lie on the same side of EF with $EG \cong ED$ and $FG \cong FD$. Proposition I.7 forces $G = D$.

Once the three vertices coincide under superposition, the corresponding angles coincide as well. This is the classical SSS argument.

3 Statement

Theorem 1 (Euclid I.8)**.** *Let A, B, C, D, E, F be points such that A, B, C are noncollinear and*

$$AB \cong DE, \qquad BC \cong EF, \qquad AC \cong DF.$$

Then

$$\angle BAC \cong \angle EDF, \qquad \angle ABC \cong \angle DEF, \qquad \angle ACB \cong \angle DFE.$$

The current Lean statement assumes noncollinearity only for ABC:

```
(hABC : not Collinear Geo A B C)
```

Noncollinearity of DEF is derived inside `HilbertSSS` from the three side congruences. The public target is exactly the conjunction of the three angle-congruence statements.

Diagrammatic audit. The preserved Book1R chapter already contained two useful classical figures: the input pair of triangles and the superposition configuration. Both are retained. What was missing was the actual geometry hidden under the public `HilbertSSS` call. Figure 3 adds that material as one two-panel formal figure.

Panel (a) records the auxiliary point G constructed on the side of AB opposite C and matched by angle and segment data to the triangle DEF. Panel (b) records the stable two-center configuration after the first SAS step: C and G lie on opposite sides of AB with $AC \cong AG$ and $BC \cong BG$. This is precisely the input to Hilbert's Theorem 17.

The internal proof of Theorem 17 itself contains further incidence cases. They are not expanded here: Theorem 17 is a reusable lower theorem called by SSS, just as the midpoint theorem was treated as a reusable component in the I.9 audit. The present figure records the proposition-relevant geometry at the level of the SSS engine, not the internal proof tree of every dependency. Thus the audited set is

$$2 \text{ classical panels} + 2 \text{ formal panels} = 4 \text{ panels in } 3 \text{ figures}.$$

4 Classical Proof

The historical dependency is simple:

$$\text{I.7} \longrightarrow \text{I.8}.$$

Superposition identifies one side of the first triangle with the corresponding side of the second. The remaining vertex is then determined by its two distances from the endpoints of the common base. Proposition I.7 supplies the uniqueness on the chosen side of that base.

This is a historical dependency. The current Lean theorem does not call `euclid_proposition_7`.

5 Proof Architecture of the Reconstruction

At the public Book I level the architecture is only

$$\begin{array}{c}\text{three side congruences}\\ \downarrow\\ \texttt{HilbertSSS}\\ \downarrow\\ \text{three angle congruences}\\ \downarrow\\ \texttt{euclid\_proposition\_8}.\end{array}$$

For audit purposes this is too compressed. The current Hilbert SSS engine has the following mathematical structure:

$$\begin{array}{c}ABC \text{ noncollinear}, \quad AB \cong DE,\ BC \cong EF,\ AC \cong DF\\ \downarrow\\ DEF \text{ noncollinear}\\ \downarrow\\ G \text{ opposite } C \text{ across } AB, \quad AG \cong DF, \quad \angle BAG \cong \angle EDF\\ \downarrow \text{ SAS on } ABG, DEF\\ BG \cong EF\\ \downarrow\\ AC \cong AG, \quad BC \cong BG\\ \downarrow \text{ Hilbert Theorem 17}\\ \angle ABC \cong \angle ABG\\ \downarrow\\ \angle ABC \cong \angle DEF\\ \downarrow \text{ rotated SAS}\\ \text{complete triangle-congruence result}\\ \downarrow\\ \texttt{euclid\_proposition\_8}.\end{array}$$

This formal proof is not Euclid's superposition argument in disguise. It is Hilbert's synthetic SSS construction organized through reusable angle, segment, separation, SAS, and rigidity theorems.

6 Hilbert Reconstruction

6.1 Derive noncollinearity of the second triangle

The API of `HilbertSSS` deliberately asks only for noncollinearity of the first triple. Its first geometric obligation is to show that the side data cannot describe a degenerate second triangle. The helper

```
hilbert_noncollinear_of_three_congruences
```

transports collinearity backwards through the three congruences and derives a contradiction with the hypothesis on ABC.

This matters for the documentation: the formal theorem is slightly stronger at the interface than a statement requiring both triangles to be declared nondegenerate by the caller.

6.2 Construct the auxiliary point on the opposite side

Let AB be the base line. The SSS proof calls the auxiliary construction

```
hilbert_sss_auxiliary_point
```

which produces G on the side of AB opposite C such that

$$AG \cong DF, \qquad \angle BAG \cong \angle EDF.$$

The helper obtains the correct side by extending through A, copying the angle by Hilbert III.4, and laying off the required segment on the resulting ray.

Using the hypothesis $AB \cong DE$, SAS for ABG and DEF then gives

$$BG \cong EF.$$

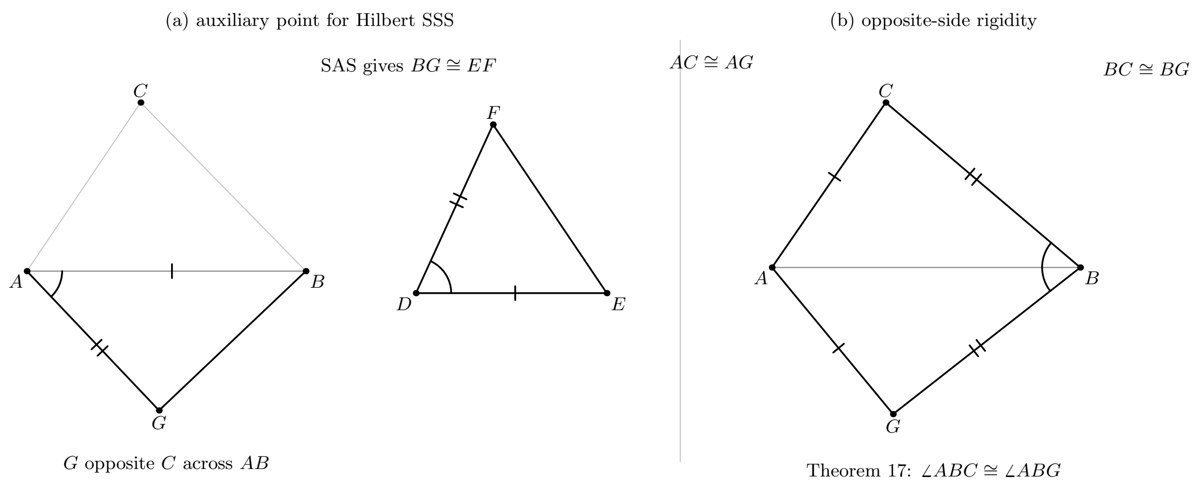

Figure 3: The hidden Hilbert SSS engine used by the current I.8 wrapper. (a) The auxiliary point G is constructed opposite C across AB and matched to DEF by $AB \cong DE$, $AG \cong DF$, and the included-angle congruence; SAS gives $BG \cong EF$. (b) Combining those results with the original SSS hypotheses gives $AC \cong AG$ and $BC \cong BG$. Hilbert's Theorem 17 converts this opposite-side two-center symmetry into the required angle rigidity at B.

6.3 Opposite-side rigidity

The original hypotheses and the first SAS result now give, by congruence transitivity,

$$AC \cong AG, \qquad BC \cong BG.$$

The points C and G lie on opposite sides of AB. Hilbert's Theorem 17 therefore applies to the two centers A, B and the two points C, G, yielding

$$\angle ABC \cong \angle ABG.$$

The first SAS comparison already gives the corresponding angle relation between ABG and DEF, so transitivity yields

$$\angle ABC \cong \angle DEF.$$

6.4 Recover the remaining angles by a rotated SAS

With the angle at B now known, the proof reverses the first side representation and applies SAS to the rotated triples

$$BAC \qquad \text{and} \qquad EDF.$$

This second SAS call returns complete congruence data. The angle at C is put into the orientation required by the public theorem using the standard angle-reversal theorem. The resulting record contains exactly the three angle fields exported by I.8.

7 Lean Formalization

The current production endpoint is intentionally short:

```
theorem euclid_proposition_8
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D E F : Geo.Point)
    (hABC : not Collinear Geo A B C)
    (hAB : Geo.Congruent A B D E)
    (hBC : Geo.Congruent B C E F)
    (hAC : Geo.Congruent A C D F) :
    Geo.AngleCongruent B A C E D F /\
    Geo.AngleCongruent A B C D E F /\
    Geo.AngleCongruent A C B D F E := by
  have hSSS :=
    HilbertSSS Geo A B C D E F hABC hAB hBC hAC
  exact
    <hSSS.2.angleA,
      hSSS.2.angleB,
      hSSS.2.angleC>
```

The shortness of this code is an API fact, not a statement that SSS is primitive. The current `HilbertSSS` theorem packages Hilbert's Theorem 18 and returns both the derived noncollinearity of DEF and a `TriangleCongruenceResult`.

There are no proposition-local helper theorems and no proposition-local case splits in `Proposition08.lean`. Source inspection of the current file finds no `sorry` and no proposition-local `axiom`. This documentation audit did not run a fresh `lake build` or `#print axioms`, so it makes no stronger compilation or axiomatic status claim.

8 Relationship with Earlier Propositions

8.1 Proposition I.7

I.7 is the decisive historical dependency. Under superposition it says that two distinct points on the same side of a fixed base cannot have the same two prescribed distances from the endpoints. The current Lean theorem I.8 does not call `euclid_proposition_7`; the Hilbert SSS theorem supplies a different rigidity proof.

8.2 Proposition I.4

I.4 and I.8 are the two early triangle-congruence landmarks of Book I. I.4 exposes SAS, while I.8 exposes SSS. In the present architecture the lower Hilbert SSS proof itself uses the reusable SAS theorem as an ingredient. This is a dependency of the Hilbert engine, not a call from I.8 to the Book I wrapper `euclid_proposition_4`.

8.3 Transition to Proposition I.9

Euclid uses I.8 immediately in I.9 to compare the two triangles surrounding the candidate angle bisector. The current I.9 implementation instead calls `HilbertSSS` directly. Thus I.8 and I.9 share the same lower triangle-rigidity interface even though the Book I call graph does not contain an edge from the formal I.9 theorem to the formal I.8 theorem.

9 Hilbert and Book Zero Components Used

9.1 Hilbert SSS

`HilbertSSS` is the direct dependency of the public theorem. It is Hilbert Theorem 18 in the project interface and is the mathematical engine whose structure is exposed in Figure 3.

9.2 Opposite-side geometry

The auxiliary point is deliberately constructed on the side of AB opposite the original third vertex. This is essential input to Hilbert Theorem 17 and is not visible in the public SSS statement.

9.3 Angle and segment construction

The helper `hilbert_sss_auxiliary_point` uses the congruence layer to copy the target angle and lay off the target side on the selected ray. The proof therefore remains synthetic: no coordinates, numerical lengths, or numerical angle measures are introduced.

9.4 SAS and Hilbert Theorem 17

The lower proof uses SAS twice. The first application determines BG in the auxiliary triangle; Theorem 17 then supplies an angle comparison across the base, and the final rotated SAS recovers the complete triangle-congruence record.

Theorem 17 is treated as a reusable lower theorem in this chapter. Its own internal incidence case split is not reproduced as proposition-specific geometry.

10 Formalization Notes

The main correction to the preserved Book1R chapter is architectural. The sentence "Proposition I.8 becomes a direct application of HilbertSSS" is correct only at the public endpoint. It is incomplete as a description of the mathematical proof because the current SSS theorem is a substantial derivation rather than a primitive axiom or an opaque declaration.

The audit therefore distinguishes three levels:

1. Euclid's historical proof: superposition plus I.7;
2. the Book I API endpoint: one call to `HilbertSSS`;
3. the hidden Hilbert engine: auxiliary opposite-side construction, SAS, Theorem 17, and rotated SAS.

The formal figure is intentionally limited to two stable states. It does not turn every theorem call into a panel, and it does not recursively expand the proof of Theorem 17. This preserves modularity while still showing the geometry that the old theorem-call diagram concealed.

11 Dependency Summary

Classical:

$$
\begin{gathered}
AB \cong DE,\ AC \cong DF,\ BC \cong EF \\
\downarrow \\
\text{superposition} \\
\downarrow \\
\text{I.7 fixes the remaining vertex} \\
\downarrow \\
\text{corresponding angles congruent} \\
\downarrow \\
\text{I.8.}
\end{gathered}
$$

Public formal endpoint:

$$\texttt{HilbertSSS} \longrightarrow \texttt{euclid\_proposition\_8}.$$

Hidden mathematical engine:

$$\begin{gathered}\text{three side congruences and } ABC \text{ noncollinear}\\ \downarrow\\ DEF \text{ noncollinear}\\ \downarrow\\ G \text{ opposite } C \text{ across } AB\\ \downarrow\\ \text{angle copy + segment layoff}\\ \downarrow\\ \mathrm{SAS}(ABG, DEF)\\ \downarrow\\ AC \cong AG,\ BC \cong BG\\ \downarrow\\ \text{Hilbert Theorem 17}\\ \downarrow\\ \angle ABC \cong \angle DEF\\ \downarrow\\ \text{rotated SAS}\\ \downarrow\\ \texttt{TriangleCongruenceResult}\\ \downarrow\\ \texttt{euclid\_proposition\_8}.\end{gathered}$$

12 Audit Status

This revision audits the documentation against the currently inspected production `Proposition08.lean`, the current Hilbert interface around `HilbertSSS`, and the preserved Book1R chapter. The production Lean files have not been modified.

All three Asymptote figure sources in the audit package were compiled during this documentation build and visually inspected after rendering. No fresh Lean compilation or axiom audit was performed as part of this documentation pass.

13 Conclusion

Proposition I.8 is a particularly clear example of the difference between a short public formal theorem and the geometry supporting its interface. Euclid proves SSS by superposition and I.7. The Book I Lean endpoint merely calls `HilbertSSS`. One level lower, however, the Hilbert proof is a substantial synthetic argument built from an opposite-side auxiliary point, SAS, two-center rigidity, and a second SAS comparison.

The audited documentation now displays all three viewpoints without confusing them. The classical proof remains historically visible, the public Lean theorem remains accurately short, and the hidden SSS engine is no longer represented as if it were a primitive black box.